Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Lab on Virtual Machines

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

Virtual Machines baseline

Virtual Machines (**VMs**) are a virtualization technology that allows developers and IT teams to run multiple operating systems on a single physical machine, each with its own virtualized hardware, providing isolated and consistent environments across different host systems.

Main Virtual Machines concepts

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- **Hypervisor**: Software that creates and manages VMs by abstracting physical hardware, e.g., Type 1 (bare-metal) or Type 2 (hosted).
- **Guest OS**: The operating system installed inside a VM, which runs independently of the host OS.
- **Isolation**: Each VM operates in a sandboxed environment, preventing interference with other VMs on the same host.
- **Resource Allocation**: VMs share physical resources (CPU, memory, storage) of the host, which can be allocated dynamically or fixed per VM.

VirtualBox

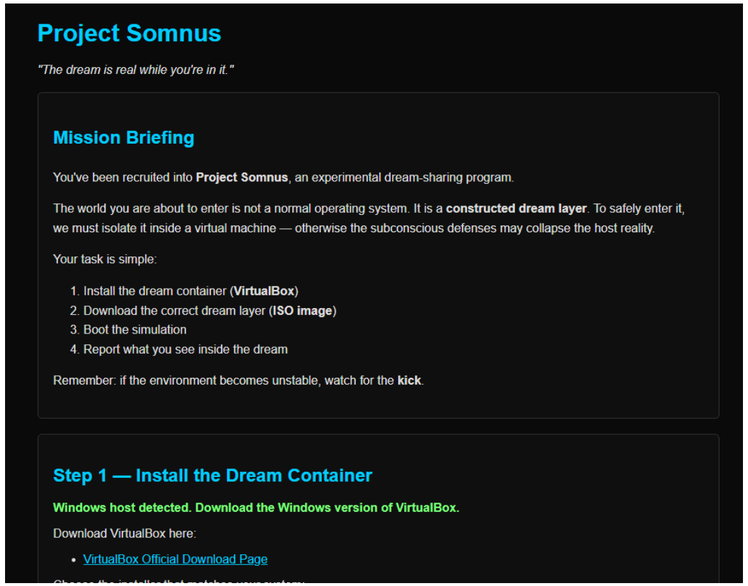
VirtualBox

- **Open Source:** The VirtualBox hypervisor is released under the GNU GPL v2 license, allowing free use, modification, and distribution.
- **Cross-Platform:** Works seamlessly on Windows, macOS, Linux, and Solaris hosts.
- **Snapshots:** Lets you save the current state of a VM and revert back anytime, making testing and experimentation safe and easy.
- **Safe to Experiment:** You can install and test other operating systems without affecting your main system or data.

MKISO

Virtual Machines Challenge 01

MKISO - Reconnaissance



Project Somnus

"The dream is real while you're in it."

Mission Briefing

You've been recruited into **Project Somnus**, an experimental dream-sharing program.

The world you are about to enter is not a normal operating system. It is a **constructed dream layer**. To safely enter it, we must isolate it inside a virtual machine — otherwise the subconscious defenses may collapse the host reality.

Your task is simple:

1. Install the dream container (**VirtualBox**)
2. Download the correct dream layer (**ISO image**)
3. Boot the simulation
4. Report what you see inside the dream

Remember: if the environment becomes unstable, watch for the **kick**.

Step 1 — Install the Dream Container

Windows host detected. Download the Windows version of VirtualBox.

Download VirtualBox here:

- VirtualBox Official Download Page

What can we observe? The challenge...

- let you download an ISO file;
- require you to install a virtual machine and retreive the flag;

MKISO - Hypothesis

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2. Keep GRUB with the default "Automated Install" option (otherwise the flag will not be in the system)

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The student needs to learn how to install an operating system:

1. Create a VM and mount the correct ISO
2. Keep GRUB with the default "Automated Install" option (otherwise the flag will not be in the system)
3. Follow the steps